Shift-Scan: Power-of-Two Decay and Ternary Projections for Multiplication-Free State Space Language Models

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Model: Mamba-2 130M · **Data:** FineWeb-Edu · **Hardware:** NVIDIA Tesla T4-class GPU (Colab / Kaggle)

Abstract

State space models (SSMs) such as Mamba keep a fixed-size recurrent state, avoiding the growing KV cache of Transformers, yet they still rely on dense multiplications in their projections and in the per-token state decay. We study whether two modifications can reduce this arithmetic cost in Mamba-2 130M while preserving language-modelling quality: (i) **ternary projections** with weights in {-1, 0, +1}, and (ii) **power-of-two decay**, where the state-decay coefficient is represented as 1 - 2^-k so that the multiplication becomes a shift. We compare three decay quantizers (Nearest PoT, HARPQ, Dyadic2) and several ternarization schemes, using quantization-aware training (QAT) with state-aware distillation, multi-seed runs, ablations, and a post-training (PTQ) validation suite on real model activations.

The main findings are: (1) naive ternarization of a pretrained model fails catastrophically (PPL of order 10^13), but a recipe with TWN quantization, a ramped schedule, learned scales and a few full-precision (FP) projections reaches **31.83 ± 0.10 PPL** over three seeds at 4M tokens (29.28 at 12M tokens), against an FP student at 23.10; (2) power-of-two decay is nearly lossless when applied post-training to the pretrained model (+0.006 to +0.022 nats), and **Dyadic2 is significantly better than Nearest PoT** (paired bootstrap, 95% CI excludes 0), while **HARPQ is statistically indistinguishable from Nearest PoT**; (3) ternarization damage is **highly concentrated**: ternarizing layer 0 alone adds 42.1 nats of cross-entropy and layer 23 adds 4.7, while most other layers add 0.1 to 0.6; (4) a custom Triton ternary kernel was about 80 times slower than FP16, and a roofline analysis shows why practical speedups are confined to bandwidth-bound decoding. We report these results, including negative ones, with explicit limitations.

Keywords: state space models, Mamba-2, ternary quantization, power-of-two quantization, quantization-aware training, efficient language models, hardware-efficient AI

1. Introduction

Transformer language models pay for context length twice: KV-cache memory grows linearly and attention computation grows quadratically. SSMs such as Mamba [1] and Mamba-2 [2] summarise the past in a recurrent state, so inference memory stays bounded. Their remaining cost is dominated by (a) large projection matrices and (b) the decay multiplication in the state update.

This motivates two opportunities. First, ternary weights can turn projection multiplications into additions, subtractions and zero-skipping, as explored for Transformers in BitNet b1.58 [3]. Second, if the decay coefficient is constrained to 1 - 2^-k, the update h_t = a·h_{t-1} + … can be implemented with a right shift in fixed-point arithmetic. The central research question is:

> **Which SSM components can be ternarized or shift-parameterized without significantly damaging language-model accuracy or long-context stability?**

1.1 Contributions

1. A QAT and state-aware distillation pipeline for ternary Mamba-2 projections, with a systematic comparison of ternarization methods and training ablations (Sections 5.2 to 5.3).
2. A controlled, training-free validation suite on the real pretrained model that compares decay quantizers with paired statistics, evaluates loss by position up to 4096 tokens, and measures per-layer sensitivity (Sections 5.5 to 5.6).
3. Evidence that ternarization failure is concentrated in a few layers, giving a data-driven basis for mixed precision (Section 5.6).
4. A candid hardware analysis showing a large gap between theoretical multiplication elimination and measured kernel speed (Section 5.8).

1.2 Scope and honesty statement

Ternary SSMs and SSM quantization are already active research areas (Section 2), so novelty is **not** claimed for applying ternary weights to Mamba. Results here come from a single 130M model, perplexity-centric evaluation and limited seeds. Two evidence tiers are kept separate throughout: **trained** (QAT) results from recovered experiment logs, and **post-training** (PTQ) results from the validation suite.

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2. Background and Related Work

Mamba and Mamba-2. Mamba introduced selective SSMs with input-dependent parameters [1]; Mamba-2 reformulated them through structured state space duality, with a scalar-per-head decay a_t = exp(-A·Δ_t) [2]. This per-head, per-token scalar decay is the quantity targeted by power-of-two parameterization here.

Low-bit language models. BitNet b1.58 shows that ternary weights trained from scratch can approach FP quality in Transformers [3]. Ternary Weight Networks (TWN) [4] choose a threshold from the weight magnitude statistics and a per-group scale, and training through quantizers commonly uses the straight-through estimator (STE) [5].

Quantization of SSMs. Quamba and Quamba2 study post-training quantization of selective SSMs, and Slender-Mamba and ternary Mamba variants explore low-bit SSM training. Because these exist, the defensible contribution of this project is the *combination* of power-of-two decay, ternary projections, controlled component-wise sensitivity, long-context analysis and practical kernel evaluation, rather than any one ingredient. (Full citation details are listed in the reference note.)

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3. Method

3.1 Ternary projections

Projection weights are quantized to W ∈ {-1, 0, +1} with scaling factors, so a projection can in principle use only additions, subtractions and zero-skipping. Four quantizers were evaluated: **absolute-mean ternarization**, **TWN** (threshold-based) [4], **activation-aware ternarization**, and scaled ternary projections, all trained with STE-based QAT [5]. A ramp schedule (gradual transition from FP to ternary), learned scales, and retention of selected sensitive projection rows in FP were studied as training options.

3.2 Power-of-two decay

The decay is approximated as â = 1 - 2^-k with integer k, so multiplication by 2^-k is a right shift. Three quantizers were compared:

Quantizer	Rule
Nearest PoT	Choose k that minimises
HARPQ (Horizon-Aware Retention Quantization)	Choose k minimising a uniform-weight, finite-horizon retention-curve MSE, mean over lags l = 1..H of (â^l - a^l)^2, with H = 128 and k in 1..14

Quantizer	Rule
Dyadic2	Represent 1-a as a two-term dyadic sum 2^-i + 2^-j (i, j in 1..16), choosing the nearest value in log(1-a)

HARPQ follows the objective recorded in the project's calibration file (`fineweb_harpq_calibration.json`: 200,192 calibration tokens, 20,000 sampled coefficients, uniform lag weights). In that calibration, HARPQ changed only 2.74% of exponents relative to Nearest PoT and reduced the objective by 0.98%, which already suggests a small effect. In the validation suite, Dyadic2 is implemented as defined above and may differ in details from the version used in the v2 training code.

3.3 Hybrid-precision design

Not every component is assumed quantizable. The components considered are projection matrices, state decay, Δ, B, C and the recurrent state. The strategy is to keep sensitive components in higher precision and lower precision only where evidence supports it (Section 5.6).

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4. Experimental Setup

Model and data. Mamba-2 130M, evaluated and trained on FineWeb-Edu (`sample-10BT`). The teacher is the pretrained FP model; the student is initialised from it.

Training (trained tier). QAT with state-aware distillation (KD loss plus state loss), sequence length 1024, FP16, group size 128, on T4-class GPUs (a full 100M-token run exceeded memory; runs used micro-batch 1 with gradient accumulation 16). Budgets: 1M, 4M, 10M, 12M and 20M tokens.

Validation suite (PTQ tier). Training-free experiments on the pretrained FP32 model using a fixed, cached FineWeb-Edu token stream (300k tokens), so every method sees identical text. The decay is intercepted by hooking the `dt` channel of each mixer's input projection, capturing the true per-token decay a = exp(-Δ·A) and replacing it with a quantized value by inverting softplus. Statistics use a **paired bootstrap** over text chunks (4,000 resamples, 95% intervals). Experiment sizes: decay comparison 64 chunks × 2048 tokens; loss-by-position 12 chunks × 4096 tokens; layer sensitivity 16 chunks × 1024 tokens.

Metrics. FineWeb-Edu perplexity (PPL), cross-entropy (CE, nats), state MSE, WikiText PPL and LAMBADA accuracy where available.

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5. Results

5.1 Pilot feasibility

Configuration	FineWeb-Edu PPL
FP decay baseline	21.025
Shift decay + FP projections	21.201
Shift decay + preliminary ternary projections	≈ 1.07 × 10^13

Shift decay alone barely changed perplexity (+0.18), but directly ternarizing the pretrained model failed catastrophically. The pilot used only 4,096 evaluation tokens and an STE-rounded `last_k` in the decay-grid analysis, so it served as a feasibility and debugging step, not validation. Section 5.6 explains the failure mechanism.

5.2 Ternarization methods (4M tokens, three seeds)

Method	Seed PPLs	Mean PPL
FP student (reference)	23.087, 23.109, 23.095	23.10
Absolute-mean ternary	60.89, 70.93, 55.35	62.39
Activation-aware ternary	48.53, 50.04, 50.48	49.69
TWN	44.10, 43.27, 45.55	**44.30**

TWN was clearly the strongest ternarizer and became the basis of the v2 configuration.

5.3 Training ablations and the v2 configuration

1M-token ablations (single run each):

Configuration	PPL
Base	996.98
Learned scale	755.11
LR 0.002	1988.30
LR 0.004	4080.94
Ramp	74.37

Configuration	PPL
FP-sensitive projections	41.51
Combined	**39.79**

High learning rates and the base setup are unstable; ramping and keeping sensitive projections in FP account for most of the recovery. This is consistent with the concentration effect found independently in Section 5.6.

v2 best configuration: TWN, learning rate 0.001, ramp 0.2, learned scaling, 6 FP-sensitive projection rows, and the original decay (no decay quantization in the selected configuration).

Budget	Seeds (PPL)	Mean ± SD
4M tokens	31.794, 31.765, 31.943	**31.834 ± 0.096**
12M tokens (single run)	29.283	29.283

Relative to the FP student (23.10), the 4M-token model is 1.38× in perplexity and the 12M-token model 1.27×, so a substantial ternarization gap remains but narrows with more training.

5.4 Decay quantization inside the trained model

Method (best config, single seed)	PPL
Original decay	31.794
Nearest PoT	31.820
Dyadic2	31.827

Differences are about 0.03 PPL, so in the trained configuration the choice of decay quantizer is negligible (single seed; no significance test possible).

5.5 Preliminary 10M and 20M-token HARPQ experiments

10M tokens, decay-only comparison (FineWeb-Edu PPL):

Seed	Nearest PoT	HARPQ	Dyadic2
0 (run log)	109.5	101.0	103.1

Seed	Nearest PoT	HARPQ	Dyadic2
1	96.08	96.30	99.74
2	117.13	113.43	114.82
Mean	107.6	103.6	105.9

Seed-to-seed variation (96 to 117 PPL) is far larger than any method difference, and the ranking is inconsistent across seeds (Nearest PoT is best in seed 1). These runs do not establish a ranking. In seed 2, HARPQ had the best FineWeb-Edu PPL, WikiText PPL (306.0) and LAMBADA accuracy (2.60%), while Dyadic2 had the lowest state MSE (0.662).

20M tokens, HARPQ, seed 0 (19,988,480 tokens, step 610): student PPL 133.9 versus teacher 24.5 (CE 4.897 vs 3.199; state MSE 0.654). This early pipeline left the student far from the teacher, motivating the v2 recipe.

5.6 Post-training validation on the real model

These experiments use the **pretrained FP32** model with quantization applied after training. They characterise the model's sensitivity; they are not QAT results.

Experiment 1: real decay traces. Per-token decay was captured from real forward passes. Mean absolute retention-product error over windows of length H:

Method	H=16	H=64	H=256	H=1024
Nearest PoT	0.00658	0.00404	0.00179	0.00130
HARPQ	0.00723	0.00439	0.00162	0.00099
Dyadic2	0.00182	0.00102	0.00039	0.00019

On the stationary retention-curve MAE (lags 1..128, 20,000 real coefficients), Nearest PoT = 0.01611, HARPQ = 0.01595 and Dyadic2 = 0.00493. These values closely reproduce the independently recorded calibration (0.01577 and 0.01563), which supports consistency between pipelines. HARPQ helps only at long horizons and slightly hurts at short ones.

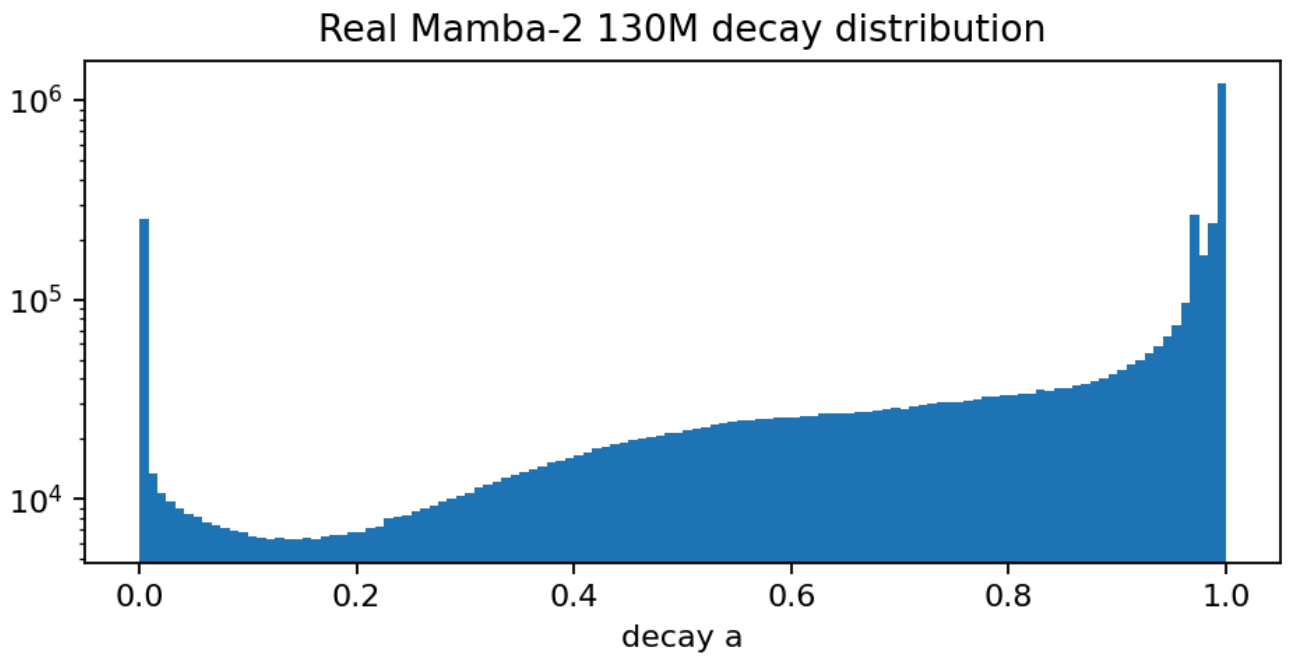


Figure 1. Histogram (log count) of per-token decay a over all 24 layers of the real Mamba-2 130M, captured from forward passes on FineWeb-Edu.

Experiment 2: decay-only perplexity with paired CIs (131k tokens).

Method	PPL	ΔCE vs FP (nats)	95% CI
FP baseline	20.215	0	n/a
Nearest PoT	20.659	+0.0218	[0.0202, 0.0234]
HARPQ	20.669	+0.0222	[0.0206, 0.0239]
Dyadic2	**20.330**	**+0.0057**	[0.0050, 0.0064]

Paired difference	Δ nats	95% CI	Verdict
HARPQ − Nearest PoT	+0.0005	[-0.0002, +0.0012]	not significant
Dyadic2 − Nearest PoT	-0.0161	[-0.0175, -0.0146]	Dyadic2 significantly better
HARPQ − Dyadic2	+0.0165	[+0.0151, +0.0180]	Dyadic2 significantly better

Power-of-two decay is nearly lossless for the pretrained model, and a two-term dyadic representation recovers about three quarters of the Nearest PoT loss. HARPQ gives no measurable advantage, matching the 1% objective reduction in its calibration file.

Experiment 3: long-context loss by position (12 × 4096 tokens).

Method	CE early (0-512)	CE late (3584-4095)	Late - early
FP	2.9995	2.8160	-0.1835
Nearest PoT	3.0193	2.8564	-0.1629
HARPQ	3.0234	2.8445	-0.1789
Dyadic2	3.0062	2.8217	-0.1845

No catastrophic error accumulation appears up to 4096 tokens. Dyadic2's gap to FP stays near 0.005 to 0.01 nats throughout; the Nearest PoT gap drifts from roughly 0.015 early to 0.04 late, so "small" is accurate but "perfectly flat" is not. With only 12 chunks, this is supporting evidence rather than proof of long-context stability.

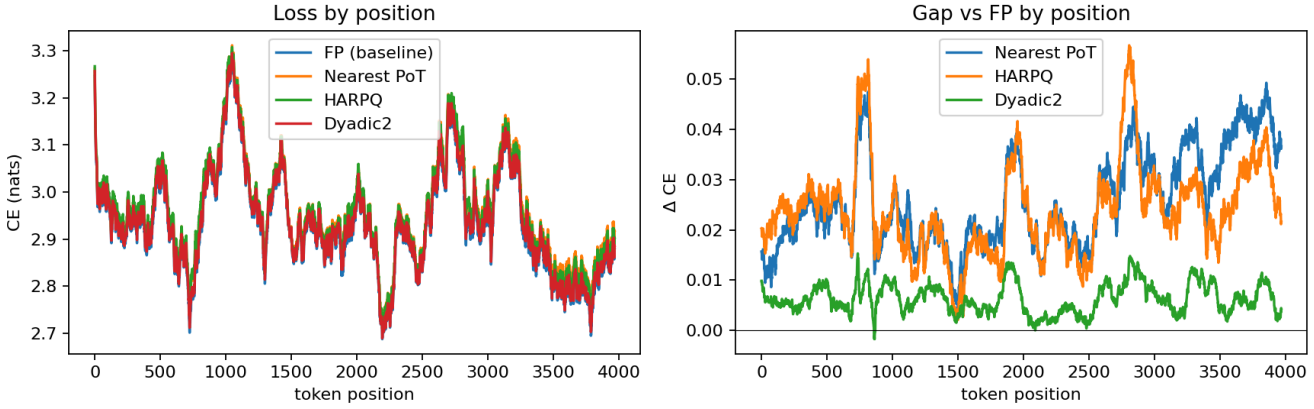


Figure 2. Cross-entropy by token position (left) and gap to the FP model (right), smoothed over 128 tokens; 12 chunks of 4096 tokens.

Experiment 4: per-layer sensitivity. Each layer is perturbed alone; values are the CE increase over the FP model (16 × 1024 tokens).

Layer	Ternarize in_proj + out_proj (TWN, PTQ)
0	**+42.15**
23	**+4.67**
3	**+1.05**
20, 13, 19, 16, 4	+0.55 to +0.62
remaining layers	+0.11 to +0.50

Decay quantization is benign in every layer: the largest single-layer effect is +0.0031 nats (layer 23, both Nearest PoT and HARPQ). Ternarization damage, in contrast, is concentrated: layer 0 alone costs 42 nats, which is sufficient to explain the 10^13

perplexity of the pilot. Layers 0, 23 and 3 are therefore natural candidates for FP retention, a data-driven refinement of the hand-chosen FP-sensitive rows in Section 5.3.

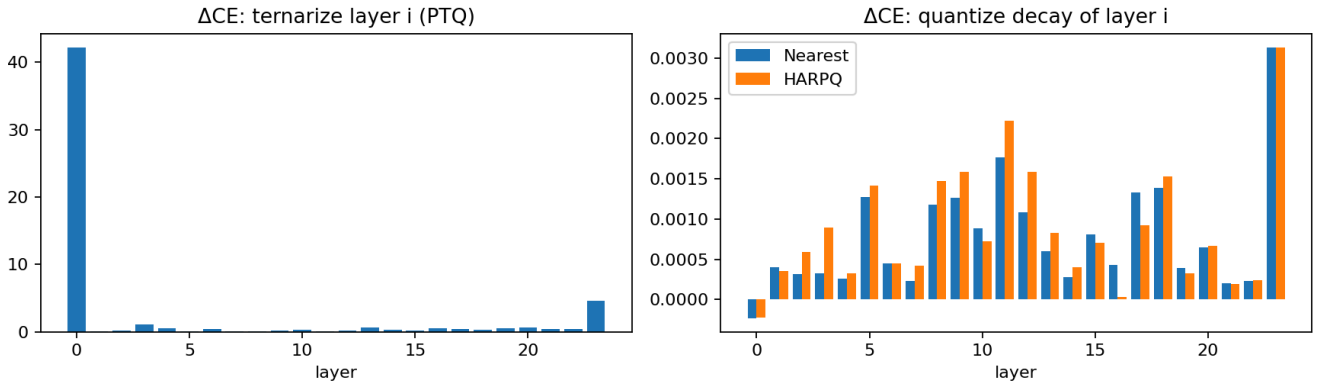


Figure 3. Cross-entropy increase when ternarizing one layer (left, post-training) and when quantizing one layer's decay (right). The y-scales differ: up to 42 nats on the left, 0.003 nats on the right.

5.7 Storage and arithmetic estimates

Configuration	Estimated compression
Ternary + 2-bit packing + FP16 group scales	2.331×
Base-3 ternary + scales	2.424×
Ternary + FP-sensitive projections	1.967×
Ternary + INT8 embedding/head	2.786×
Ternary + INT4 embedding/head	3.496×

These are analytical estimates and were not all verified on deployed checkpoints. An operation census estimates about **85.0%** multiplication elimination within SSM layers but only **62.1%** model-wide, because the FP16 language-model head accounts for about 71.1% of the remaining multiplications. Projection quantization alone therefore cannot remove model-level multiplication cost.

5.8 Hardware: kernel and roofline analysis

A custom Triton ternary projection kernel matched FP16 numerically (maximum relative error ≈ 0.0005) but was far slower: **12.98 ms versus 0.159 ms at M = 512, a 0.0123× speedup**. No inference speedup can be claimed from this kernel.

A roofline measurement of an FP16 projection (layer-0 `in_proj`) explains the structure of the opportunity:

Batch rows M	FP16 time (ms)	Effective bandwidth (GB/s)	Ideal 2-bit speedup
1	0.0246	209	7.9×
16	0.0270	191	7.9×
128	0.0405	127	none (compute-bound)
512	0.1038	50	none (compute-bound)

At small M the layer is memory-bound, so a perfect 2-bit kernel could in principle approach the byte ratio (about 7.9×). At M ⩾ 128 FP16 tensor cores dominate and no ternary gain is expected. The T4 has no ternary arithmetic, so realising gains requires bit-packed, lookup-table or popcount-style kernels rather than masked accumulation. Timing varied between sessions (about 142 to 209 GB/s at M = 1), so these figures should be read as indicative.

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6. Discussion

Decay is the easy part; projections are the hard part. Constraining the decay to a power of two costs 0.006 to 0.022 nats post-training and about 0.03 PPL in the trained model, whereas ternarizing projections costs 8 to 17 PPL even with the best recipe. Reducing arithmetic cost in SSMs will therefore depend mainly on the projection side.

Concentrated fragility explains training behaviour. The 42-nat effect of layer 0, together with layers 23 and 3, accounts for the pilot failure, for why ramping and FP-sensitive rows rescued training, and for why a uniform ternarization recipe underperforms. Mixed precision guided by measured sensitivity is a principled alternative to hand selection.

HARPQ's null result is informative. Under its own objective HARPQ changes about 3% of exponents and barely moves the objective, and neither the real-trace error, the decay-only perplexity nor the long-context profile shows a significant benefit over Nearest PoT. A nonstationary CPU study on synthetic traces motivated horizon-aware quantization, but synthetic traces are not measured Mamba decays, and the real-data evidence does not support a claim of superiority.

Theory versus hardware. Eliminating 85% of SSM-layer multiplications does not yield speed on commodity GPUs: the model-wide figure is 62.1%, the naive kernel is two orders of magnitude slower, and the ideal bound applies only to bandwidth-bound decoding.

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7. Limitations and Threats to Validity

- **Accuracy gap.** The best ternary student (31.83 PPL at 4M tokens; 29.28 at 12M) remains above the FP student (23.10).
- **Evidence tiers.** Decay and sensitivity findings (Section 5.6) are post-training on the pretrained model; they do not replace trained-model ablations.
- **Statistical power.** Several trained results are single-seed (12M run, decay comparison in Section 5.4, the 1M ablations); the 10M comparison is dominated by seed variance.
- **Single model and metric.** Only Mamba-2 130M and mainly FineWeb-Edu perplexity; benchmark coverage (WikiText, LAMBADA) is incomplete, and LAMBADA accuracies are very low at this scale.
- **Long context.** Evaluation reaches 4096 tokens on 12 chunks; stability beyond this is untested.
- **Hardware.** No optimised shift-add kernel, no latency, throughput or energy measurement beyond the roofline estimate.
- **Implementation fidelity.** Dyadic2 in the validation suite is an implementation of the stated rule and may differ from the v2 training implementation.
- **Reproducibility.** Some original checkpoints, optimizer states and tokenizer caches were lost (Section 9); recovered metrics are preserved with provenance but cannot recreate missing artifacts.

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8. Future Work

Phase	Plan
A. Fair baselines	FP16 projections with original decay, Nearest PoT, HARPQ and Dyadic2, matched tokens, data and seeds
B. Combined	Ternary projections with each decay quantizer, three seeds
C. Sensitivity-guided mixed precision	Keep layers 0, 23 and 3 in FP, ternarize the rest with QAT; extend sensitivity to Δ, B, C and the state
D. Long context	Sequences beyond 4096 with loss-by-position, state error and retention error
E. Hardware	Bit-packed or LUT-based kernels; latency, throughput, memory and energy per token
F. Dissemination	Public repository and paper with full ablations

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9. Reproducibility and Data Provenance

A data-loss event removed some checkpoints and experiment records. The recovered package contains the original 28-run `runs.jsonl`, supplemental metrics extracted from logs and tables, configuration, README and provenance notes; the original records are preserved unmodified. The validation suite is a single Colab notebook that uses a fixed cached token stream, saves `results.json`, per-experiment CSVs and figures (decay histogram, loss-by-position, layer sensitivity), and reproduces the FP baseline PPL (20.2147) exactly across independent runs, which was used as a sanity check on the pipeline. Results labelled as estimates (Section 5.7) are analytical.

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10. Conclusion

Shift-Scan shows that, for Mamba-2 130M, power-of-two decay is nearly free, with a two-term dyadic form significantly more accurate than nearest rounding, while ternarizing projections is the real difficulty and its damage is concentrated in a few layers, most strikingly layer 0. A TWN-based QAT recipe with ramping, learned scales and FP-sensitive projections reduces the pilot's catastrophic failure to a 1.27 to 1.38× perplexity gap relative to an FP student. HARPQ, as defined, offers no measurable gain over Nearest PoT, and the

custom kernel demonstrates that arithmetic savings do not automatically become speed. The most promising next step is sensitivity-guided mixed precision with controlled multi-seed ablations and a bandwidth-aware kernel.

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Reference note: Quamba, Quamba2, Slender-Mamba and ternary Mamba works are discussed in Section 2 from the project's literature review; their full bibliographic details (authors, venues, identifiers) should be verified and added before external submission.